

Core Concepts

Differential Equations - Mathematics

1. The central results

This week presents the principal results of the module and the conditions under which they hold. Each is stated formally before being applied, since the conditions are what determine whether a result may be used in a given situation.

2. Applying the results

Worked examples show the standard method of application, including the checks that must be performed before a theorem may be invoked. Skipping those checks is the most common source of error in assessed work.

3. Common mistakes

Misapplying a result outside its hypotheses, and confusing a necessary condition with a sufficient one, account for most lost marks. Both are avoided by returning to the formal statement rather than working from memory of the example.